Project 1

The first step , we have to import CSV file into pandas library so that it can read data and analyse data. Pandas is a powerful data analysis library providing flexible way to do data manipulation. It helps in organizing, cleaning, and analysing data using Data Frames and Series.

In order to load the CSV file using Pandas, we need to do the following step.

```
#Load CSV file
df=pd.read_csv(r"C:\Users\Windows\Downloads\railway_gauges 1.csv")
print(df.head())
```

Pandas returns as Data frame object. So here we have used 'df.head()' function so that it give a first 5 rows and columns of the dataset. If we want last 5 rows and columns we can use 'df.tail()' function.

Output:

```
      Year  Broad Gauge  Metre Gauge  Narrow Gauge  Total
0  1964-65         3521         2891           464   6876
1  1965-66         3603         2913           470   6986
2  1966-67         3615         2911           481   7007
3  1967-68         3601         2920           506   7027
4  1968-69         3603         2921           508   7032
```

Next we need to find at which year the railway tracks are installed on more number so to find out that we can out idxmax() function.

```
#Find which year had the maximum installations
max=df.iloc[[df["Total"].idxmax()]]
print(max)
```

Here are the pandas functions we used:

- **idxmax()**: gives the index of the max value
- **iloc[]**-is primarily an integer based location search — iloc[3]
- **loc[]-**is primarily a label based location search — loc['Broad Gauge']

Output:

```
=====================================================================
       Year  Broad Gauge  Metre Gauge  Narrow Gauge  Total
48  2012-13         6577          379           216   7172
=====================================================================
```

In 2012-13, we have a greater number of railway track installations.

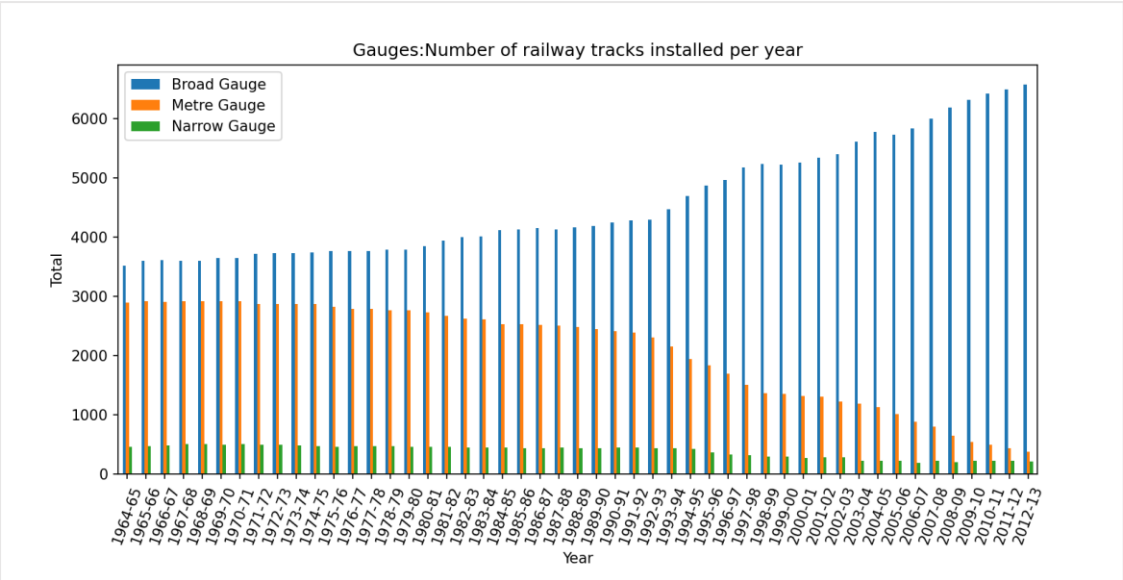
Data Visualisation

Matplotlib is a plotting library used to create graphs and visualizations like line charts, bar charts, and histograms.

Below steps are used to see the data using bar chart.

```
#plot data using bar chart
df=df.drop('Total',axis=1)
ax=df.plot(x="Year",kind="bar")
plt.xticks(rotation=70)
plt.xlabel('Year')
plt.ylabel('Total')
plt.title('Gauges:Number of railway tracks installed per year')
plt.show()
```

Here are the some functions we used:

- **df.drop():** is used to remove rows or columns from a Data Frame.
- **df.plot():** Creates a line graph by default.
- **plt.xticks():** is used to get or set the tick locations and labels on the x-axis of a plot.
- **plt.xlabel():** is used to label the x-axis of plot.
- **plt.ylabel():** is used to label the y-axis of plot.
- **plt.title():** is used to add a title to your plot**.**
- **plt.show():** is used to display the plot window**.**

Output: